Java Assignment: Abstract Class - Shape Calculator

Objective

Understand abstract classes, abstract methods, inheritance, method overriding, runtime polymorphism, and arrays of objects in Java.

Problem Statement

Create an abstract class **Shape** with:
• Variable: color
• Abstract methods: area(), perimeter()
• Concrete method: displayColor()

Create subclasses **Circle**, **Rectangle**, and **Triangle**. Each subclass must implement the abstract methods using the appropriate mathematical formulas.

Requirements

1. Accept required dimensions from the user.
2. Store all shape objects in an array of Shape.
3. Display the color of each shape.
4. Calculate and display area and perimeter for every object.
5. Use runtime polymorphism while invoking overridden methods.
6. Do not use Collections or Exception Handling.

Expected Output

For every shape, display:
• Shape Type
• Color
• Area
• Perimeter

Submission Guidelines

Submit source code with proper comments and meaningful class, method, and variable names.